

Financial Deserts: A Subnational Atlas of Financial Exclusion

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May 2026

Abstract

Over 4,700 municipalities across Peru, Brazil, Colombia, and the Philippines—a third of all subnational units in the four countries—have zero bank offices. Using administrative data from financial regulators covering 10,214 districts and municipalities across Latin America and Asia from 2002 to 2025, we construct the first subnational comparative atlas of physical financial access and document three facts. First, financial deserts are pervasive: 73% of Peruvian districts, 46% of Brazilian municipalities, 25% of Colombian municipalities, and 32% of Philippine municipalities lack any bank office as of 2024. Second, deserts are structural, not transitional: 96% of Peruvian and 64% of Philippine deserts persist over a decade. Third, physical banking infrastructure is contracting across all three Latin American countries (15–24% decline from peaks), while the Philippines shows the opposite—offices doubling from 6,000 to 13,000—yet deserts persist there too. These patterns suggest that neither digital finance expansion (LAC) nor physical office growth (Philippines) automatically resolves the structural exclusion of peripheral territories.

Keywords: financial inclusion, financial deserts, territorial inequality, bank branches, developing countries

JEL Codes: G21, O16, R12

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1 Introduction

Financial inclusion is widely recognized as a determinant of economic development and a dimension of inequality in its own right (??). Yet the dominant measures of financial access—the World Bank’s Global Findex and the IMF’s Financial Access Survey—operate at the national level, obscuring enormous subnational variation. A country may report 70% account ownership nationally while entire regions lack any financial infrastructure. This aggregation masks precisely the spatial inequalities that matter most for policy.

The concept of “deserts”—territories defined by the complete absence of a critical service—has been productive in public health, where “food deserts” identify areas lacking grocery stores and fresh food (?). We apply this logic to finance. A **financial desert** is a subnational unit with zero bank offices. The term captures a qualitative threshold: where physical financial infrastructure is entirely absent, residents cannot originate credit, access formal savings instruments, purchase insurance, or receive financial advisory—services that remain incompletely substitutable by digital channels or correspondent agents (??).

This paper introduces the Financial Deserts Atlas, the first comparative subnational database of physical financial access in developing countries. Covering Peru, Brazil, Colombia, and the Philippines from 2002 to 2025, the atlas documents three facts. First, financial deserts are pervasive: a third of all subnational units across the four countries lack any bank office. Second, deserts are persistent: once a territory becomes a desert, it almost never gains a bank office. Third, the three Latin American countries are experiencing a contraction in physical banking infrastructure, while the Philippines shows the opposite—rapid office expansion that still fails to eliminate deserts. In both cases, peripheral territories remain structurally excluded.

These findings speak to a growing literature on the spatial determinants of financial access. ? showed that India’s social banking program, which mandated branch openings in unbanked locations, reduced rural poverty—establishing that physical bank presence has causal effects on economic outcomes. ? found similar impacts from Banco Azteca’s expansion into underserved Mexican municipalities. ? demonstrated that branch supply affects household financial inclusion even in high-income economies. Our contribution is to document that, across Latin America, the supply of physical banking infrastructure is not merely insufficient but actively declining.

The remainder of the paper proceeds as follows. Section ?? describes the data sources and methodology. Section ?? presents our results on the magnitude, persistence, population distribution, and temporal dynamics of financial deserts. Section ?? discusses the implications of these patterns. Section ?? concludes.

2 Data and Methodology

We use administrative records from three national financial regulators to construct annual panels of bank office counts at the finest available subnational level.

For **Peru**, the Superintendencia de Banca, Seguros y AFP (SBS) provides the number of offices by institution type per district, annually from 2002 to 2025. Institution types include commercial banks (Banca Múltiple), municipal savings banks (Cajas Municipales), rural savings banks (Cajas Rurales), EDPYMES, and finance companies. Peru’s administrative structure comprises 1,891 districts, with a median population of 3,090 (INEI Census, 2017). The panel is balanced: $1,891 \text{ districts} \times 24 \text{ years}$.

For **Brazil**, the Banco Central do Brasil (BCB) publishes ESTBAN (Estatística Bancária por Município), a regulatory filing that records which bank agencies report deposits in each municipality each month. We use December snapshots from 2002 to 2024, accessed via basedosdados.org. ESTBAN captures both openings and closures: an agency that stops reporting has effectively closed. Brazil has 5,570 municipalities; the panel is balanced ($5,570 \times 23 \text{ years}$). Population data come from the IBGE Census (2022).

For **Colombia**, the Superintendencia Financiera (SFC) reports the number of offices per entity per municipality, quarterly from 2008 to 2025, available through the national open data portal (datos.gov.co). Colombia has 1,121 municipalities in our sample. Municipalities absent from SFC filings in a given year are coded as having zero offices. Population data come from the DANE Census (2018).

For the **Philippines**, the Bangko Sentral ng Pilipinas (BSP) publishes the “Geographical Distribution of Banking Offices” at the city/municipality level, annually from 2012 to 2024, available through the BSP statistics portal. The data cover universal/commercial banks, thrift banks, and rural/cooperative banks. The Philippines has 1,632 municipalities in our 2024 sample. We use December snapshots for each year. Municipalities are disambiguated by province to handle the prevalence of duplicate names (e.g., there are nine municipalities named “San Isidro” across different provinces). Population data come from the PSA Census (2020).

We define a **financial desert** as a subnational unit with zero bank offices in a given year. This definition captures access to full-service financial intermediation—credit origination, savings products, insurance, and advisory—which requires physical presence and is not fully substitutable by digital channels or agent correspondents (??). We refer to this as a “Layer 1” definition, acknowledging that additional layers of access—correspondent agents, ATMs, mobile wallets—exist but serve distinct and more limited functions.

To complement the unit-based desert rate, we compute population-weighted desert rates using fixed census populations, which measure the share of the national population re-

siding in financial deserts. This avoids conflating demographic change with changes in financial infrastructure and provides a more policy-relevant metric of exclusion.

Figure ?? illustrates the spatial distribution of financial deserts in Peru as of 2024, showing the concentration of deserts in Andean and Amazonian districts.



Figure 1: Financial deserts in Peru, Brazil, Colombia, and the Philippines, 2024. Red-shaded units have zero bank offices. Financial deserts concentrate in Andean highlands, Amazonian lowlands, Brazil’s interior Northeast, and the Philippine archipelago’s smaller island municipalities. An interactive version with year selection is available at <https://josemendozas.github.io/findeserts/>.

3 Results

Table ?? presents the cross-country snapshot for 2024. Over 4,700 subnational units across the four countries have zero bank offices. Peru exhibits the highest desert rate at 72.6%, reflecting both the granularity of its administrative divisions and the extreme concentration of banking in coastal cities. Brazil’s 46.1% covers a far larger population base: over 2,500 municipalities without a single bank agency. The Philippines, despite doubling its office count since 2012, still has 32% of municipalities unserved. Colombia, with a denser banking network relative to its administrative geography, has one in four municipalities unserved.

Table 1: Financial Deserts Across Four Countries, 2024

Country	Units	With Office	Deserts	% Desert	Offices
Peru	1,891	518	1,373	72.6%	3,986
Brazil	5,570	3,004	2,566	46.1%	8,311
Colombia	1,121	840	281	25.1%	5,425
Philippines	1,632	1,109	523	32.0%	13,318
Total	10,214	5,471	4,743	46.4%	31,040

These aggregate numbers, however, mask sharply different trajectories. Figure ?? shows the evolution of desert rates over time. Peru experienced a rapid reduction in financial deserts during the 2000s, driven by the expansion of Cajas Municipales and microfinance institutions: the desert rate fell from 92% in 2002 to 75% by 2012, then stagnated. Brazil’s trajectory is more alarming: a U-shaped curve. Brazilian financial deserts declined from 42% in 2002 to a minimum of 34% in 2013, then reversed, climbing back to 46% by 2024 as bank agencies closed faster than new ones opened. Colombia has remained stable at approximately 25% throughout 2008–2025. The Philippines presents a contrasting case: desert rates declined from 38% in 2012 to 30% in 2021, alongside a doubling of total offices (6,375 to 13,087). Yet progress stalled after 2021, and by 2024 the desert rate had risen slightly to 32%. Even rapid office expansion has limits when the remaining unserved municipalities are small, remote, and archipelagic.

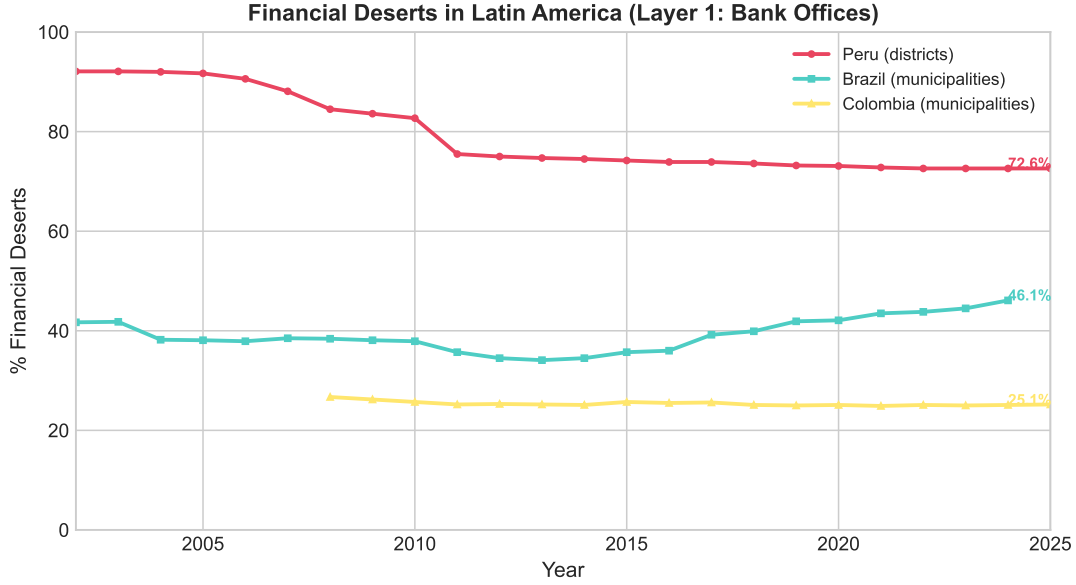


Figure 2: Percentage of subnational units classified as financial deserts, 2002–2025.

The persistence of financial deserts is striking. Table ?? reports transition matrices over the longest available window for each country. In Peru, 96% of districts that were deserts in 2012 remain deserts in 2024—only 53 districts gained a bank office over 12 years. In Brazil, persistence is lower (84.3%), but the direction of change compounds the concern: 607 municipalities *lost* their last bank agency between 2002 and 2024, substantially exceeding the 365 that gained one. Colombia’s persistence rate of 88.6% is intermediate. The Philippines, despite its rapid office expansion, still shows 64.4% persistence: nearly two-thirds of municipalities that were deserts in 2012 remain so in 2024. This suggests that new offices concentrate in already-served areas rather than reaching the most excluded municipalities.

Table 2: Persistence of Financial Desert Status

Country	Period	Desert→Desert	Desert→Access	Access→Desert	Persistence
Peru	2012–2024	1,366	53	7	96.3%
Brazil	2002–2024	1,959	365	607	84.3%
Colombia	2008–2024	265	34	16	88.6%
Philippines	2012–2024	404	223	119	64.4%

This persistence raises a fundamental question: are financial deserts a consequence of insufficient demand, or do they reflect coordination failures in which the absence of financial infrastructure depresses the economic activity that would justify entry? The literature on bank branching suggests the latter. ? found that mandated branch openings in unbanked Indian locations generated sufficient deposits to sustain operations, implying that latent demand existed but could not attract voluntary entry. The structural persistence we

document—96% over 12 years in Peru—is consistent with an equilibrium trap in which low financial access and low economic complexity reinforce each other.

The unit-based desert rate, however, overstates the share of the population affected, because financial deserts tend to be small, sparsely populated territories. Figure ?? compares the unit-based and population-weighted desert rates for Peru and Brazil. While 72.6% of Peruvian districts are deserts, only 21% of the population lives in them. The gap reflects the success of Cajas Municipales and microfinance institutions in reaching district capitals with concentrated populations after 2005, even as the majority of districts—small, dispersed, and commercially unviable—remained unserved.

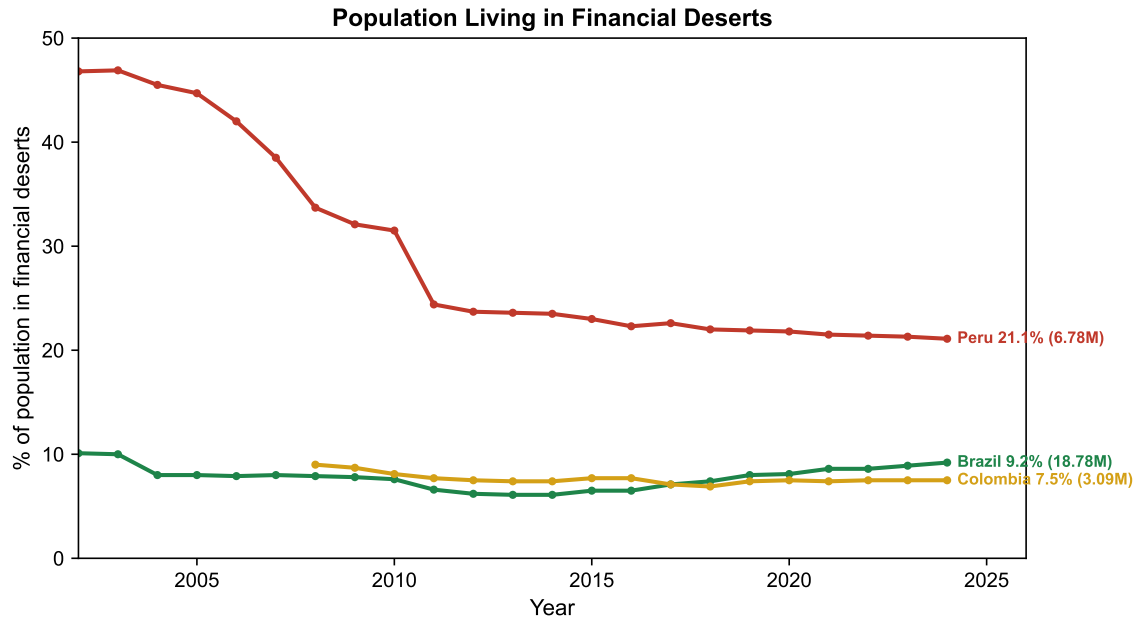


Figure 3: Share of national population living in financial deserts, Peru, Brazil, and Colombia. The population-weighted rate is substantially lower than the unit-based rate, but still affects approximately 30 million people across the three LAC countries. Philippines population data are pending integration.

In Brazil, the population story reinforces the reversal narrative. The share of the population in deserts fell to 6.1% (12.4 million) in 2013 but has risen back to 9.2% (18.8 million) by 2024. Nearly 19 million Brazilians—nearly the population of Chile—now live in municipalities without a single bank agency. This is not a statistical artifact of administrative boundaries; it represents a large and growing population excluded from physical financial intermediation.

Table 3: Population in Financial Deserts

Country	Census Pop.	Pop. in Deserts	% in Deserts	Desert Median Pop.
Peru	32.2M	6.8M	21.1%	3,090
Brazil	203.1M	18.8M	9.2%	5,737
Colombia	41.0M	3.1M	7.5%	4,850

The final dimension of our results concerns the dynamics of physical banking infrastructure. The three Latin American countries have experienced declining bank office counts since peaks reached in 2014–2019, while the Philippines shows the opposite trend. Figure ?? and Table ?? document these divergent trajectories.

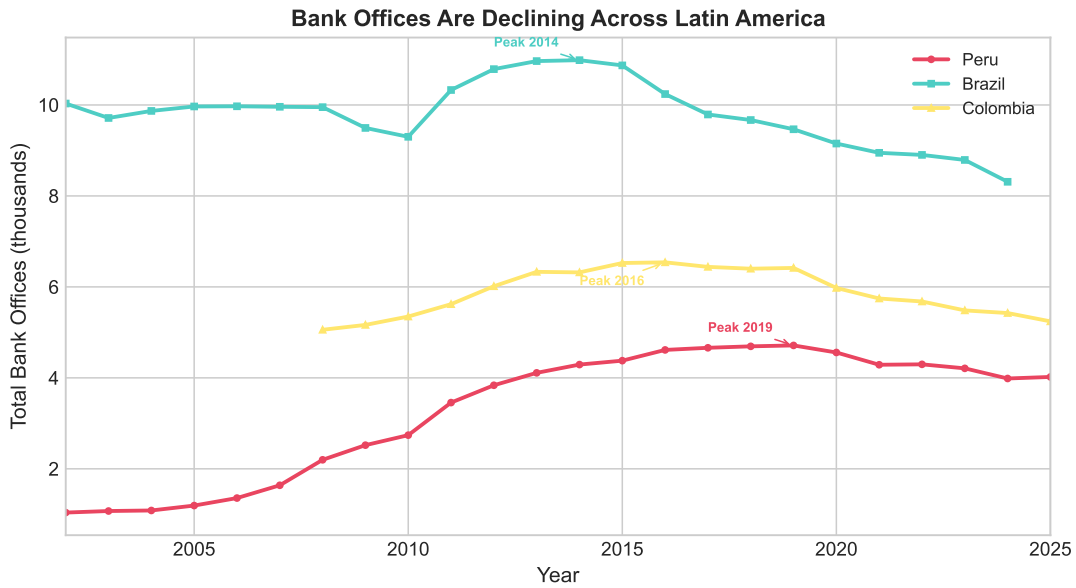


Figure 4: Total bank offices by country, 2002–2025. The three LAC countries show declining office counts while the Philippines shows sustained growth—yet deserts persist in both contexts.

Table 4: Bank Office Dynamics

Country	Peak Year	Peak Offices	2024 Offices	Change
Peru	2019	4,713	3,986	−15.4%
Brazil	2014	10,985	8,311	−24.3%
Colombia	2016	6,540	5,425	−17.0%
Philippines	2024	13,318	13,318	+108.9% ^a

^a Growth from 2012 base of 6,375 offices.

Brazil’s decline is the most severe: nearly one in four bank agencies has closed since 2014, coinciding with the rapid adoption of PIX (instant payments, launched 2020) and mobile

banking. Peru’s decline accelerated after 2019, alongside the expansion of Yape and Plin. Colombia shows a 17.0% reduction since 2016. Crucially, these closures concentrate in municipalities that *already had offices*, not in financial deserts. In Brazil, the 607 municipalities that lost their last bank agency between 2002 and 2024 outnumber the 365 deserts that gained one, producing a net increase in the number of financial deserts despite two decades of financial inclusion policy.

The Philippines provides an instructive contrast. Unlike the LAC countries, the Philippines more than doubled its banking offices between 2012 and 2024 (from 6,375 to 13,318). Yet even this rapid expansion has not eliminated financial deserts: 32% of municipalities remain unserved, and 64% of 2012 deserts persist. This suggests that market forces alone—even in a growing banking sector—do not resolve the exclusion of peripheral, low-density, or geographically fragmented territories.

4 Discussion

The patterns documented across Latin America and Asia raise a question that existing literature has not adequately addressed: why do financial deserts persist regardless of whether banking systems are expanding (Philippines) or contracting (LAC)? Is the exclusion of peripheral territories an efficient market outcome, or does it represent a structural failure?

The efficiency interpretation holds that digital channels—mobile wallets, agent correspondents, instant payment systems—can substitute for physical bank offices, making brick-and-mortar infrastructure redundant. Under this view, the decline in office counts reflects rational cost optimization, and the affected populations are adequately served through alternative channels. There is evidence for this in urban and peri-urban contexts, where digital adoption is high and financial literacy sufficient for self-service (??).

The market-failure interpretation, however, is more consistent with the patterns we document. If digital channels were adequate substitutes, we would expect desert rates to decline as offices close—that is, digital access would compensate for physical withdrawal. Instead, we observe stable or increasing desert rates alongside office contraction. This suggests that the populations in financial deserts are not effectively reached by digital alternatives. The persistence of deserts—96% over 12 years in Peru—indicates that private entry does not occur even as potential markets grow. This is consistent with coordination failures of the type identified by ?: without an initial anchor institution, the fixed costs of entry exceed the expected returns from thin local markets, even though aggregate welfare gains from access may be substantial (?).

The distinction between transactional inclusion and intermediation inclusion is critical

here (??). Peru has over 166,000 correspondent agent points (agentes) reaching 99% of districts, and mobile wallets like Yape have achieved near-universal urban penetration. By transactional metrics—the ability to send and receive payments—Peru’s financial inclusion has improved dramatically. But correspondent agents cannot originate loans, evaluate credit risk, or provide financial advisory. The closure of bank offices eliminates the local capacity for intermediation, potentially locking peripheral economies into a low-complexity equilibrium in which financial services are limited to cash-in/cash-out transactions (?).

The regional implications are significant. Financial deserts in all three countries are not randomly distributed: they concentrate in territories that are also disadvantaged along other dimensions—indigenous populations, limited connectivity, dependence on subsistence agriculture (?). The retreat of physical banking from these territories risks compounding existing spatial inequalities, creating a two-tier financial system in which metropolitan populations access the full range of intermediation services while peripheral populations access only basic payment functionality (?).

Our analysis has several limitations. The administrative level differs across countries: Peru’s 1,891 districts are more granular than Brazil’s 5,570 municipalities, Colombia’s 1,121, or the Philippines’ 1,632, which mechanically affects cross-country comparisons. The definition of financial deserts as “zero offices” is binary and does not capture variation in the quality or range of services. The Philippines data start in 2012 (vs. 2002 for Peru and Brazil), limiting temporal comparisons. Finally, we do not model the determinants of desert status, which would require a different empirical framework.

5 Conclusion

The evidence from four countries across two continents suggests that financial deserts are a structural feature of developing economies that neither digital finance expansion (LAC) nor physical office growth (Philippines) automatically resolves. The persistence of deserts across such different contexts—96% in Peru, 64% in the Philippines—points to coordination failures in which peripheral territories remain trapped below the threshold of commercial viability. Future work should examine whether regulatory interventions—branching mandates, differentiated licensing, public banking models—can break these traps. The atlas, available at <https://josemendozas.github.io/findedeserts/>, provides the empirical foundation for this research agenda, with planned expansion to 20+ countries across Latin America, Asia, and Africa.

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